The property of ergodicity combined with associated mixing properties similar to those mentioned in Chap. 5, provide a framework to give precise answers to some of these questions

In the previous section we introduced the Fokker-Planck equation as a means of analyzing the evolution of the phase space density. If the system is ergodic and mixing, we expect that the phase space distribution is, ultimately, consistent with the Gibbs-Boltzmann ensemble in the sense that, for a C_0^∞ test function ϕ (smooth and compactly supported),

$$\lim_{t\to\infty}\int_{\mathcal{D}}\phi(\boldsymbol{q},\boldsymbol{p})\rho(\boldsymbol{q},\boldsymbol{p},t)\mathrm{d}\omega=\int_{\mathcal{D}}\phi(\boldsymbol{q},\boldsymbol{p})\rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega,$$

regardless of the initial distribution.

There is a key difference between the concept of microcanonical ergodicity and canonical ergodicity. For deterministic dynamics, the distribution solves the Liouville equation

$$\frac{\partial\rho}{\partial t}=-\nabla\cdot(\boldsymbol{f}\rho),\qquad\nabla=(\nabla_{\boldsymbol{q}},\nabla_{\boldsymbol{p}}).$$

In the case of an SDE, the distribution may spread out over time. Even a single phase space point, which corresponds to a delta-function initial distribution, will instantly diffuse as the solution of the Fokker-Planck equation and, eventually, would cover the entire phase space. The incorporation of randomness has clearly changed the picture of the evolving phase space distribution in a fundamental way.

We have already seen a canonically ergodic SDE in a simplified setting in the example of the Ornstein-Uhlenbeck equation for a single degree of freedom (6.20), given an initial point distribution. The solution $p(t)$ to the Ornstein-Uhlenbeck equation, for any finite time t, is always normally distributed

$$\rho(p,t)=\exp\left(-\frac{(p-\mu(t))^2}{2\sigma(t)^2}\right),\tag{6.44}$$

where

$$\mu(t)=p_0e^{-\gamma t},\qquad\sigma^2(t)=\mathrm{k_B}\mathrm{T}m\left(1-e^{-2\gamma t}\right).\tag{6.45}$$

It is easy to see that the distribution converges to a unique invariant distribution (in this case a Gaussian with mean zero and variance $\mathrm{k_B}\mathrm{T}m$ as $t\to\infty$). Ergodicity is easily verified in the case of the Ornstein-Uhlenbeck equation, since we can find its exact distributional solution. However, we can still demonstrate ergodicity for more complex SDEs where it is extremely challenging (or impossible) to find a solution or even to give qualitative statements about the solution.